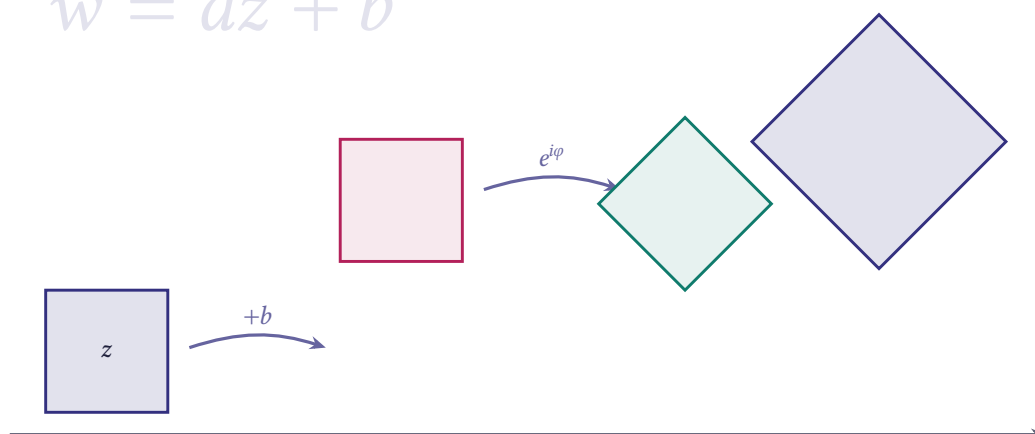


Linear Mappings

نگاشت‌های خطی

$$w = az + b$$



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This chapter examines some simple mappings from the domain plane (z) to the range plane (w).

4.1 Translation

$$\begin{aligned} w = z + b &= (x + iy) + \operatorname{Re}\{b\} + i\operatorname{Im}\{b\} \\ &= (x + \operatorname{Re}\{b\}) + i(y + \operatorname{Im}\{b\}) \end{aligned}$$

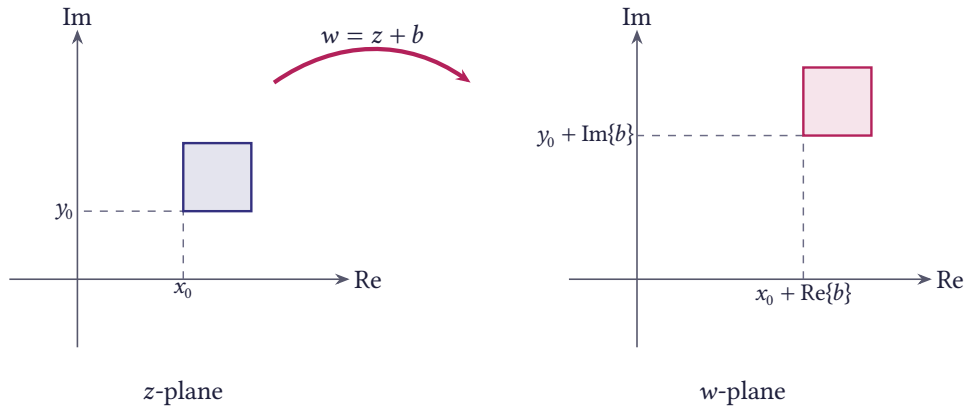


Figure 4.1 Translation $w = z + b$: a region of the z -plane is shifted by b without changing its shape

4.2 Rotation

$$w = az, \quad |a| = 1, \quad a = e^{i\varphi}, \quad \varphi = \arg\{a\}$$

$$z = |z|e^{i\theta} \implies w = |z|e^{i(\theta+\varphi)}$$

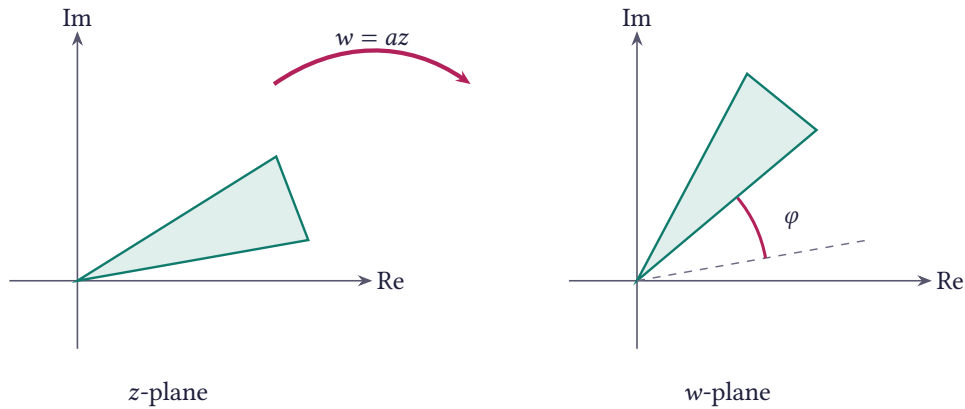


Figure 4.2 Rotation $w = az$ with $|a| = 1$: a region of the z -plane is rotated about the origin by $\varphi = \arg a$

4.3 Dilation or Contraction

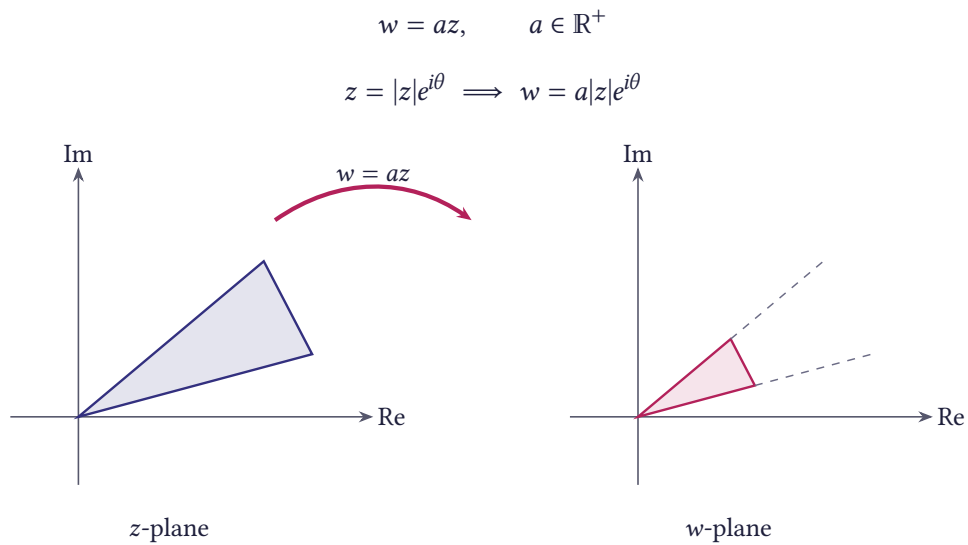


Figure 4.3 Contraction $w = az$ with $0 < a < 1$; for $a > 1$ the mapping is a dilation

4.4 The General Linear Mapping

$$w = az + b, \quad a, b \in \mathbb{C}$$

- (a) a rotation by $\arg a$
- (b) a dilation or contraction by $|a|$
- (c) a translation by b

Example 4.1

For $w = (1 + i)z + (1 - i)$, find the image of the unit square.

$$w(0) = 1 - i, \quad w(1) = 2, \quad w(i) = 0, \quad w(1 + i) = 1 + i$$

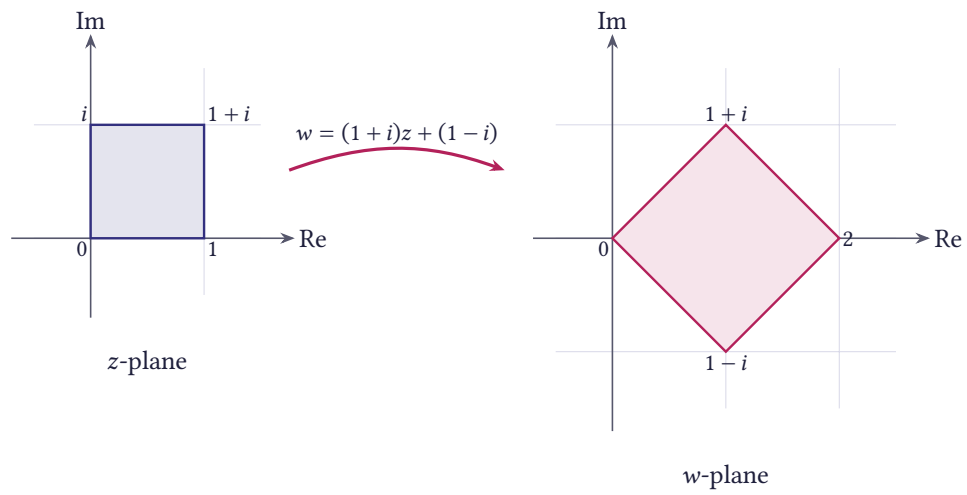


Figure 4.4 The image of the unit square under the mapping $w = (1+i)z + (1-i)$